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Markdown

7) Feature Transformation (if needed)

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- The process of applying mathematical or statistical transformations to the existing features in a dataset to make them more suitable for a machine learning algorithm or to reveal underlying patterns in the data.
- Feature transformation techniques aim to improve the quality and representativeness of the features, which can lead to better model performance and more meaningful insights.

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Python

```python
# Normalization
scaler = MinMaxScaler()
normalized_data = scaler.fit_transform(data)

# Standardization
scaler = StandardScaler()
standardized_data = scaler.fit_transform(data)

# Logarithmic Transformation
log_transformed_feature = np.log(data['feature1'])

# Power Transformation
power_transformed_feature = np.sqrt(data['feature2'])

# Box-Cox Transformation
boxcox_transformed_feature, _ = boxcox(data['feature3'])

# Binning
bin_edges = [0, 20, 40, 60]
binned_feature = pd.cut(data['feature1'], bins=bin_edges, labels=False)

# Polynomial Transformation
polynomial_features = pd.DataFrame({
    'feature1_squared': data['feature1'] ** 2,
    'feature1_cubed': data['feature1'] ** 3
```